

Solutions to Chapter 3 Exercises:

1. Metric Selection in Intrusion Detection

- To minimize false alarms → **maximize specificity** or **precision**.
- When missing attacks is unacceptable → **maximize recall** or **sensitivity**.
- A balanced metric → **F1-score**, combining precision and recall.

2. Streaming Data Preprocessing

Key behaviors:

- Scaling: running mean and variance updated per batch.
- Missing values: imputed using online statistics (e.g., incremental mean).
- Incremental PCA: updates eigenvectors using partial fits rather than recomputing from scratch.

As batches arrive:

- Distributions shift (e.g., due to traffic bursts or congestion).
- PCA components drift to reflect evolving data patterns.
- Variance ratio stabilizes over time.

3. Softmax Interpretation

- Softmax converts logits into normalized probabilities that sum to 1.
- Similar traffic types produce close logits → ambiguous probabilities.
- Miscalibrated softmax may:
 - overconfidently misclassify critical traffic,
 - misprioritize flows (e.g., starving latency-sensitive traffic),
 - cause incorrect firewall decisions.

4. Sampling in Networking Datasets

- **Random sampling**: uniform representation, may miss rare events.
- **Systematic sampling**: picks every k-th entry, risks periodic bias.
- **Stratified sampling**: ensures representation of rare events → **best for anomalies**.

5. Correctness Analysis

Issues and fixes:

- Duplicate timestamps → remove or average together.
- Negative packet sizes → replace with correct entries or discard.
- Missing protocol fields → impute (mode) or mark as "unknown."

Impact:

- Removes noise and errors → improves model accuracy and reduces bias.

6. Feature Engineering for Wireless Link Prediction

Useful engineered features:

- RSSI variance (mobility indicator).
- Moving average of SNR.
- Retransmission rate.
- Link stability score (ratio of successes to attempts).
- Doppler mobility estimate (from device speed).

Each helps capture mobility, interference, and fading dynamics.

7. Incremental PCA

- Updates components using partial SVD per batch, instead of full matrix decomposition.
- Explained variance grows then stabilizes as more data arrives.
- Preferred because it uses low memory and supports streaming.

8. Cross-Validation

- Fold accuracies vary depending on class imbalance and randomness.
- Mean accuracy → indicator of expected performance.
- Variance → stability; low variance implies strong generalization.

9. Regularization in Delay Prediction

- **L1**: best when features are sparse or irrelevant (feature selection).
- **L2**: best when features are noisy or correlated (shrinks them smoothly).
- **Elastic Net**: best when both sparsity and correlation are present.

10. L1 Regularization for Feature Selection

L1 logistic regression results:

- Some coefficients → driven to exactly **zero**, meaning uninformative features.
- Retained features often include:
 - flow duration,
 - TCP flag patterns,
 - average packet size,
 - byte/packet ratios,
 - inter-arrival time statistics.

